EFFECT OF AFICAMTEN ON PATIENT-REPORTED HEALTH STATUS IN NON-OBSTRUCTIVE HYPERTROPHIC CARDIOMYOPATHY

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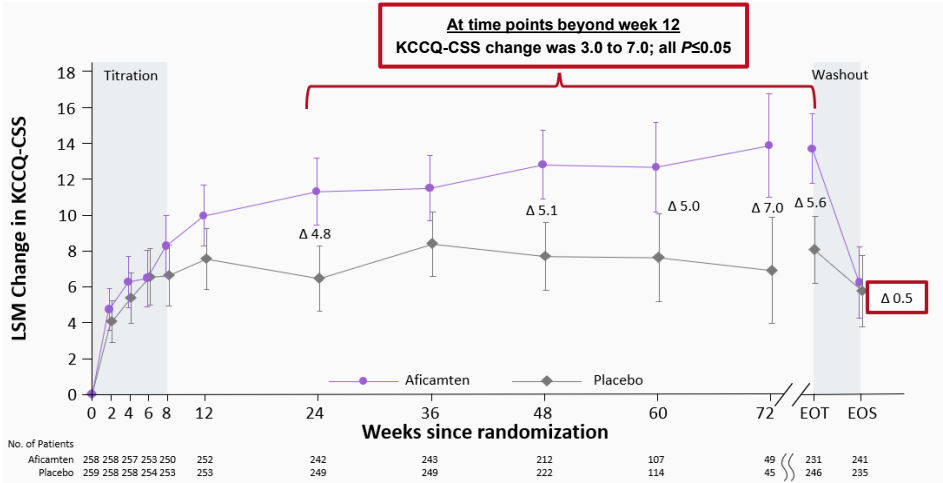
DISCLOSURES

Michael E. Nassif, MD

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BACKGROUND

- Non-obstructive hypertrophic cardiomyopathy (nHCM) can significantly impair patients' health status; their symptoms, function, and quality of life.[1,2]
- No therapies have been proven to improve patients' health status in nHCM.
- Aficamten is a next-in-class cardiac myosin inhibitor approved in oHCM.[3]



KCCQ-CSS, Kansas City Cardiomyopathy Questionnaire–Clinical Summary Score; LSM, least squares mean; nHCM, nonobstructive hypertrophic cardiomyopathy; oHCM, obstructive hypertrophic cardiomyopathy; pVO_2, peak oxygen uptake. **1.** Zaiser E, et al. *J Patient Rep Outcomes* 2020;4(1):102. **2.** Ommen SR, et al. *J Am Coll Cardiol* 2024;83(23):2324-2405. **3.** Myqorzo (aficamten) prescribing information. FDA; 2026.

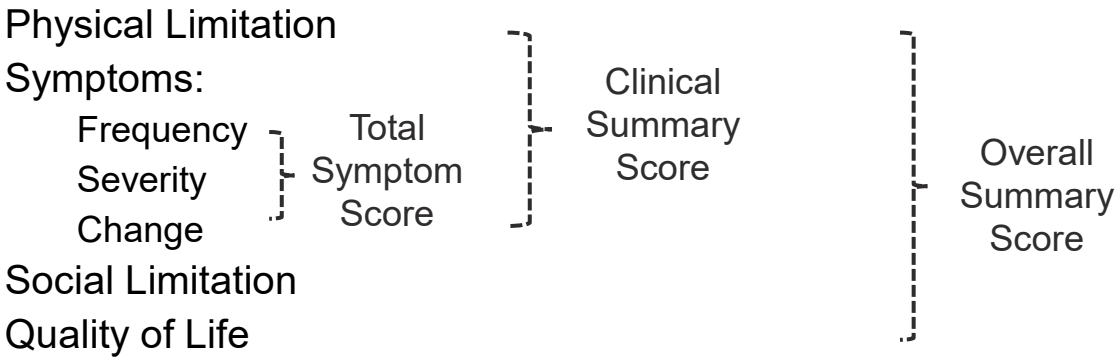
STUDY DESIGN

- International, double-blind, randomized trial of aficamten vs. placebo.
- 160 sites, 517 patients
- Key Inclusion Criteria: LV wall thickness >15mm (>13 with family history); LVOT gradient <30mmHg at rest, <50mmHg with Valsalva; EF>65%; KCCQ-CSS<85.
- Primary Outcomes: 36-week Change in KCCQ-CSS and pVO_2.
- After last participant completed 36-week follow-up and ≥200 had completed 1 year, the trial was stopped. An End of Trial (EOT) health status assessment was completed, blinded medications were stopped, and a 4-week End of Study (EOS) visit was performed (washout).
- KCCQ and Seattle Angina Questionnaire collected serially throughout the study.

KCCQ, Kansas City Cardiomyopathy Questionnaire; LVOT, left ventricular outflow tract; pVO_2, peak oxygen uptake

WHAT IS THE KCCQ?

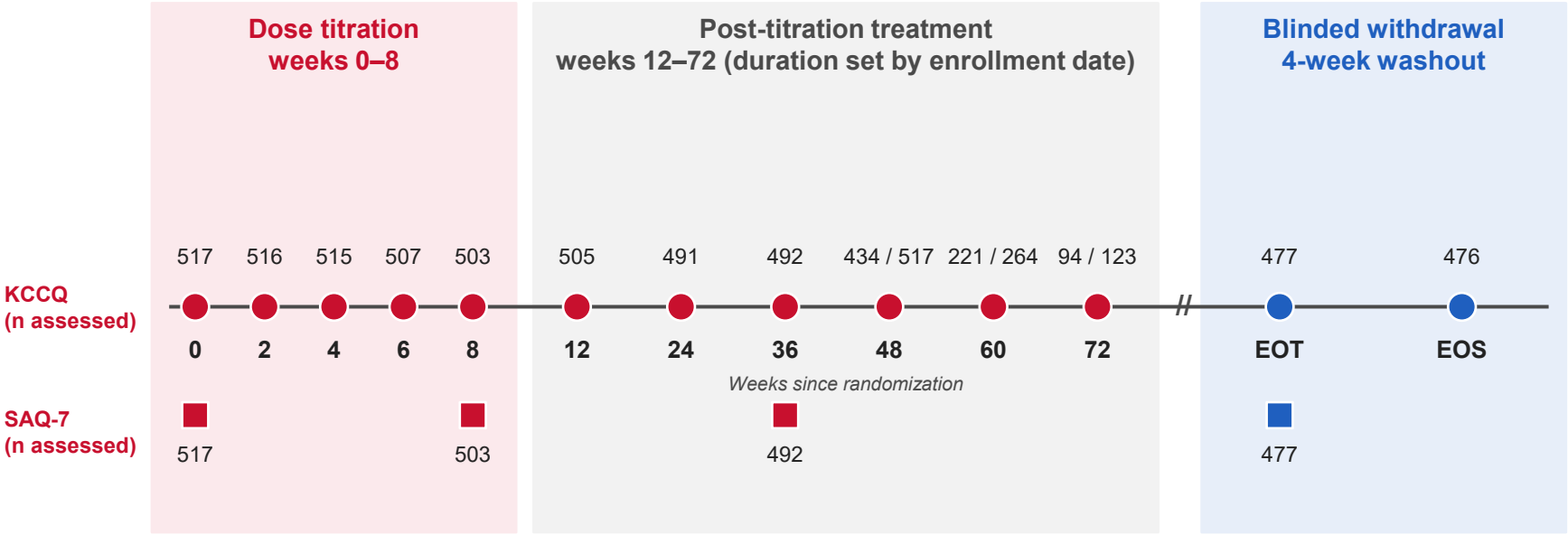
- 23 items that measure clinically relevant domains



- Represents *patients'* perspectives of their disease
- Established validity, reliability and responsiveness
  - Changes of 5, 10, 15, and 20 = small, moderate, large, and very large clinical changes
- Qualified by FDA's CDRH and CDER as a COA

CDRH, Center for Devices and Radiological Health; CDER, Center for Drug Evaluation and Research; COA, Clinical Outcomes Assessment.

METHODS: ASSESSMENT SCHEDULE AND ANALYSIS



Mixed-effects model for repeated measures (weeks 2–72): treatment × visit, stratification factors, baseline score as covariate → visit-specific differences and average treatment effects for titration, post-titration, and all visits

EOT and EOS: timing varies by participant, so analyzed separately by linear regression (EOT→EOS change adjusted for EOT score)

EOS, end of study; EOT, end of treatment; KCCQ, Kansas City Cardiomyopathy Questionnaire; SAQ, Seattle Angina Questionnaire

STATISTICAL METHODS

- **Intention to treat**
  - Inclusion criterion: Baseline and ≥1 follow-up KCCQ (98.7%);
  - Primary Outcome: KCCQ-OSS
  - Secondary Outcomes: KCCQ Subscales and SAQ
- **Repeated-measures mixed model (MMRM)** with treatment, visit, treatment × visit, strata, baseline score effects to generate:
  - Adjusted mean change and between-group difference at each visit
  - Time course: average effect by period, growth model
- **Clinical meaning:** Categorical 36-week: ≤-5, -5-<5, 5-<10, 10-<15, 15-<20, ≥20)
- **Washout:** EOT-to-EOS change to define impact of withdrawing treatment

EOS, end of study; EOT, end of treatment; KCCQ, Kansas City Cardiomyopathy Questionnaire; OSS, Overall Summary Score; SAQ, Seattle Angina Questionnaire

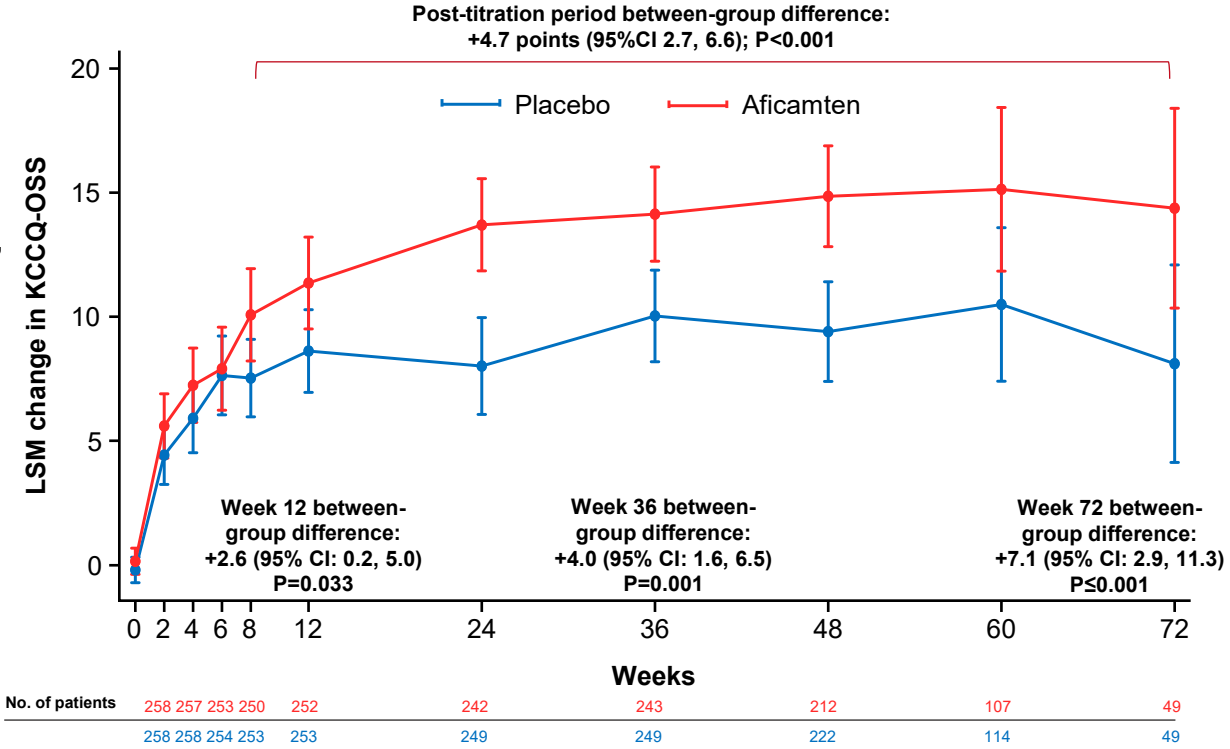
RESULTS: STUDY POPULATION AND BASELINE HEALTH STATUS

Characteristic	Aficamten n=258	Placebo n=259
Age, years	55.7 ± 16.2	54.5 ± 16.0
Female, n (%)	140 (54.3)	137 (52.9)
NYHA class III, n (%)	84 (32.6)	74 (28.6)
Background HCM therapy, n (%)		
Beta-blocker, n (%)	203 (78.7)	197 (76.1)
Non-DHP CCB, n (%)	31 (12.0)	39 (15.1)
KCCQ-OSS[a]	60.2 ± 16.6	58.8 ± 16.2
KCCQ-CSS	66.3 ± 16.2	65.0 ± 15.4
KCCQ Total Symptom	67.6 ± 17.9	66.8 ± 17.0
KCCQ Physical Limitation	65.2 ± 17.8	63.1 ± 17.5
KCCQ Social Limitation	60.0 ± 24.0	58.8 ± 21.9
KCCQ Quality of Life	48.0 ± 19.8	46.6 ± 20.8
SAQ-SS	69.3 ± 18.8	67.8 ± 19.0

Values are mean ± SD unless otherwise indicated. KCCQ and SAQ scores range from 0 to 100; higher scores indicate better health status.
HCM, hypertrophic cardiomyopathy; KCCQ, Kansas City Cardiomyopathy Questionnaire; KCCQ-CSS, Kansas City Cardiomyopathy Questionnaire-Clinical Summary Score; KCCQ-OSS, Kansas City Cardiomyopathy Questionnaire–Overall Summary Score; Non-DHP CCB, nondihydropyridine calcium channel blocker; NYHA, New York Heart Association; SAQ, Seattle Angina Questionnaire.

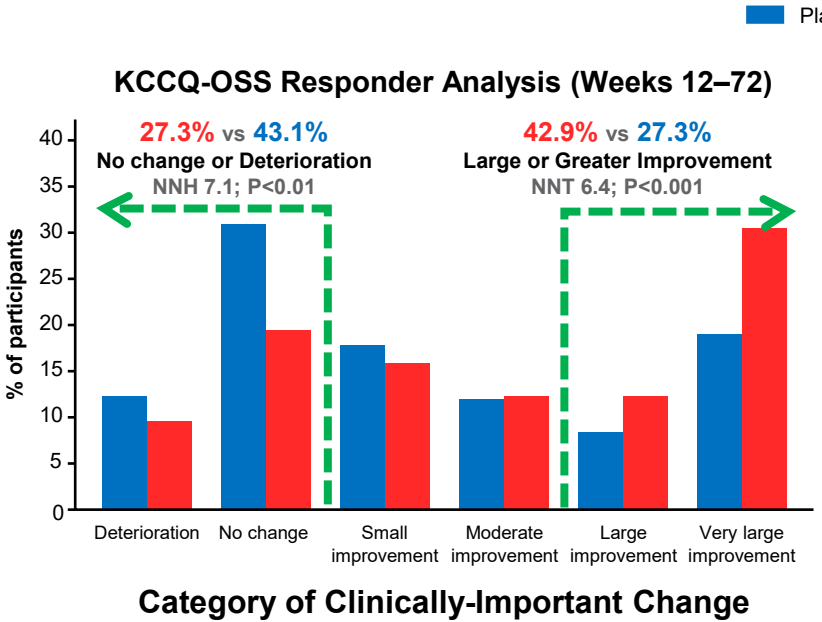
RESULTS: KCCQ-OSS TRAJECTORY

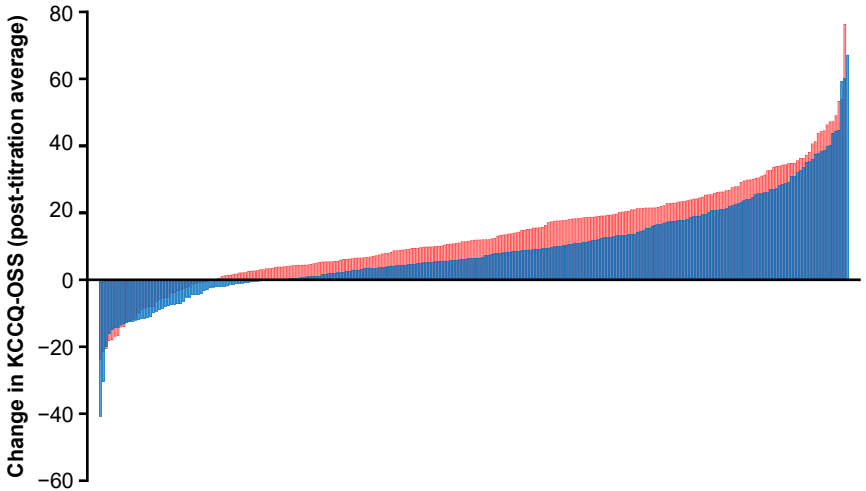
- **Non-linear Treatment Effects**
- **Titration (Weeks 2–8):** both groups improved; between-group difference +1.3 (95% CI: –0.7, 3.3); P=0.20.
- **Post-titration (Weeks 12–72):** aficamten kept improving while placebo plateaued; average difference +4.7 points.
  - The difference grew over time: +2.9 points/year (P=0.012).



Differences are aficamten minus placebo. Later samples are smaller by design. Trajectory analyses are supportive; secondary comparisons are unadjusted for multiplicity.
KCCQ-OSS, Kansas City Cardiomyopathy Questionnaire–Overall Summary Score; LSM, least squares mean

INDIVIDUAL'S CHANGES AT 36 WEEKS





Response thresholds were derived from heart failure and oHCM populations and have not been directly established in nHCM.
KCCQ-OSS, Kansas City Cardiomyopathy Questionnaire–Overall Summary Score; nHCM, nonobstructive hypertrophic cardiomyopathy; NNT number needed to treat; oHCM, obstructive hypertrophic cardiomyopathy

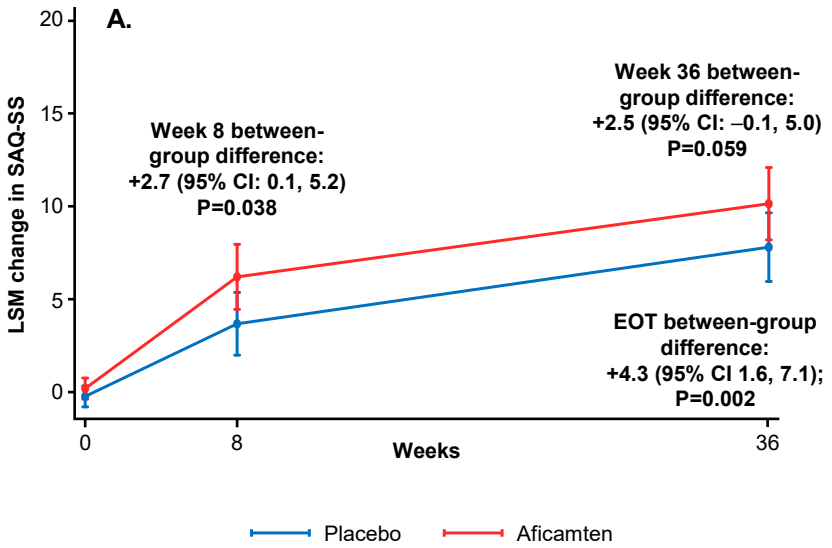
RESULTS: KCCQ SUMMARY SCORES AND DOMAINS

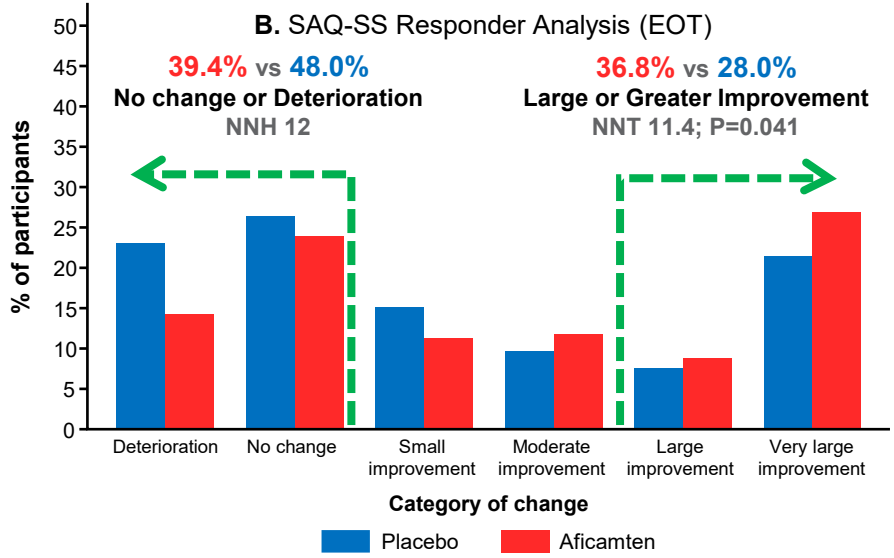
KCCQ score	Week 36 adjusted between-group difference, points (95% CI)	P value	EOT adjusted between-group difference, points (95% CI)	P value
OSS	+4.0 (1.6, 6.5)	0.001	+6.5 (3.7, 9.3)	<0.001
Total symptoms	+1.9 (–0.5, 4.4)	0.12	+5.0 (2.2, 7.7)	<0.001
Physical limitation	+4.4 (1.7, 7.1)	0.001	+6.2 (3.1, 9.2)	<0.001
QoL	+5.8 (2.7, 8.9)	<0.001	+9.4 (6.0, 12.9)	<0.001
Social limitation	+4.2 (1.0, 7.4)	0.009	+5.6 (2.1, 9.2)	0.002

Compared with placebo, aficamten improved all 4 KCCQ domains by EOT, with the largest estimated improvements for physical limitation and QoL, with the latter domain being the most impaired at baseline

Differences are aficamten minus placebo. EOT is the last on-treatment visit (week 36, 48, 60, or 72). Secondary comparisons are unadjusted for multiplicity.
CSS, Clinical Summary Score; EOT, end of treatment, KCCQ, Kansas City Cardiomyopathy Questionnaire; OSS, overall summary score; QoL, quality of life.

RESULTS: SAQ-SS



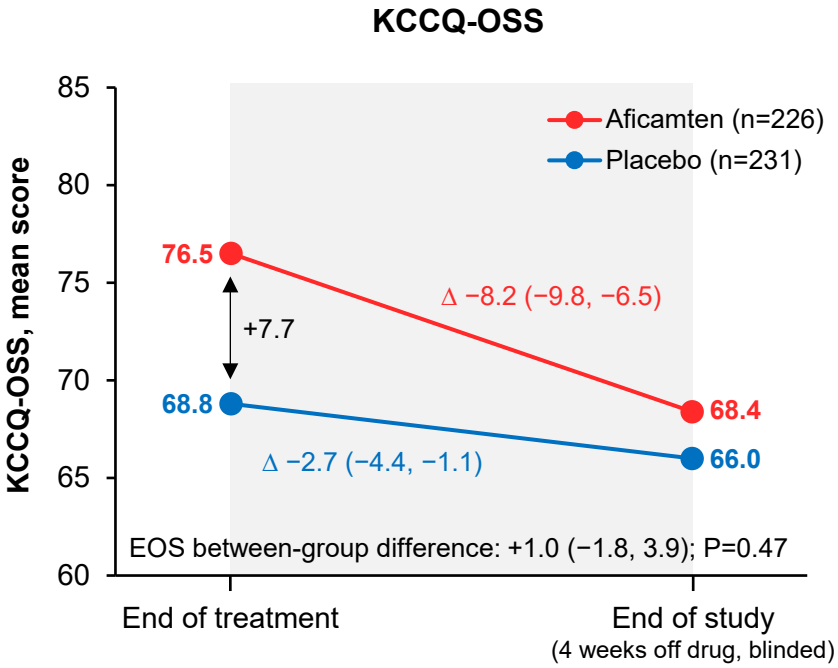


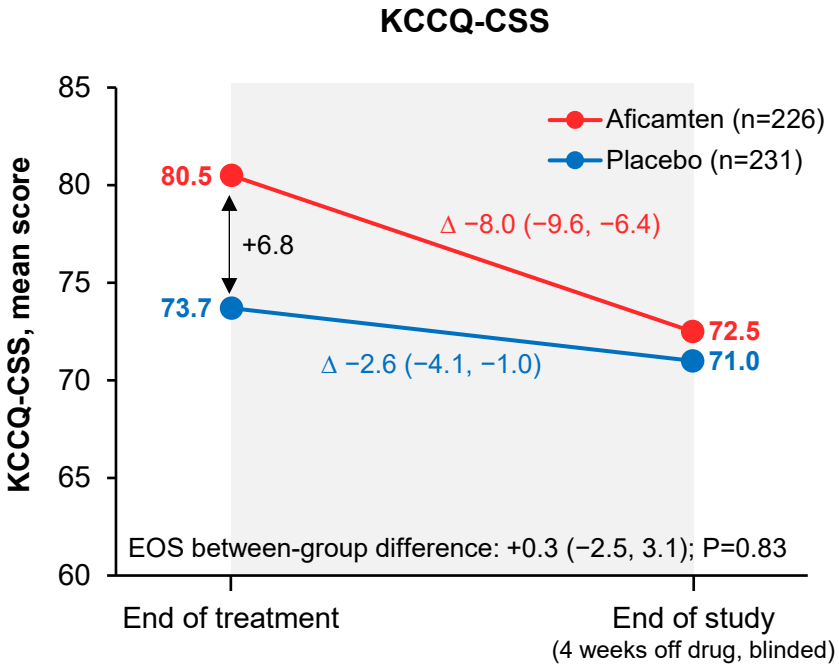
- SAQ-SS improved in both groups, with significantly greater improvement with aficamten vs placebo at Week 8 and EOT (**Figure A**).
- At EOT, more aficamten-treated vs placebo-treated participants had very large improvements (27.7% vs 20.7%) and fewer had deterioration (14.7% vs 22.4%) in SAQ-SS; NNT to prevent worsening 13.1 (95% CI: 6.9, 140.4); P=0.031 (**Figure B**).
- At Week 36, differences in the distribution of SAQ-SS change were smaller and not statistically significant (any improvement of ≥5 points: 58.8% vs 50.2%).

Differences are aficamten minus placebo. EOT varies from Week 36 to 72. Secondary comparisons are unadjusted for multiplicity; response thresholds are extrapolated.
EOT, end of treatment; LSM, least squares mean; NNT, number needed to treat; SAQ-SS, Seattle Angina Questionnaire–Summary Score.

RESULTS: TREATMENT WITHDRAWAL

Health status after blinded withdrawal of study drug (planned completers)





Includes data from planned completers with paired EOT and EOS assessments. Differences are aficamten minus placebo. Secondary comparisons are unadjusted for multiplicity.
EOS, end of study; EOT, end of treatment, KCCQ-CSS, Kansas City Cardiomyopathy Questionnaire–Clinical Summary Score; KCCQ-OSS, Kansas City Cardiomyopathy Questionnaire–Overall Summary Score..

LIMITATIONS

- **Attrition at Later Visits:** Weeks 48–72 reflect smaller, earlier cohorts
- **Response Thresholds:** Derived from HF and obstructive HCM; nonobstructive HCM anchor validation to be reported separately.
- **Selection Bias:** Enrolled high-risk cohorts (impaired status, elevated NT-proBNP); findings may not generalize to milder disease.

HCM, hypertrophic cardiomyopathy; HF, heart failure; NT-proBNP, N-terminal Pro-B-type natriuretic peptide

CONCLUSIONS

- In participants with symptomatic nHCM, aficamten improves health status vs placebo.
  - All domains - Symptoms, Physical Limitation, QoL, and Social Function improved
  - Benefits emerge after dose titration (+4.7 points post-titration)
    - Sustained through 72 weeks.
  - More patients with very large improvement; fewer deteriorate
  - Largely reversed within 4 weeks of treatment withdrawal
- Because benefits increase after titration, this should be explained to patients
- Loss of benefit after blinded withdrawal shows benefit is contingent on continued treatment.

nHCM, nonobstructive hypertrophic cardiomyopathy; QoL, quality of life.

ACKNOWLEDGMENTS

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THANK YOU

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